

MASHUK ELAHI

Electrical Engineer | Marine · Offshore · Energy · Semiconductor · Industrial

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SUMMARY

Electrical Engineer with 16 years delivering mission-critical LV/MV/HV power systems across FPSO topsides, jack-up rigs, offshore wind substations, hybrid fast crew boats, Crew Transfer Vessels (CTVs), Ro-Pax ferries, autonomous-electric tugs and 300 mm semiconductor cleanrooms. Signature strengths: first-pass class approvals across ABS-, BV-, LR- and DNV-classed projects; hybrid Li-ion BESS architecture (72 kWh commissioned, up to 50% fuel reduction); and full concept-to-handover lifecycle ownership from Pre-FEED through FEED, Detailed Engineering, EPC execution, commissioning and class interface. Certified Lead Auditor ISO 9001/50001. Advancing IACS E26/E27 OT cybersecurity, shore power (IEC/ISO/IEEE 80005-1) and DC/LVDC zonal distribution for the decarbonised, digitally governed fleet.

CAREER HIGHLIGHTS

↓**50% fuel burn** — architected 72 kWh Li-ion BESS hybrid drivetrain on a 42 m Gen-4 Fast Crew Boat; secured first-pass class approval with LR Hybrid Power notation, cutting fuel consumption by up to 50% in low-load/loitering modes.

100% IEC 61850 conformity — energised an offshore wind substation topside to full protection-relay and alarm validation before sail-away at Keppel FELS; zero punch-list rework.

Industry first — delivered Malaysia's first solar-wind autonomous unmanned wellhead platform, eliminating routine diesel use and enabling full remote operation.

↓**50% on-site labour** — prefabricated power skids for a 300 mm semiconductor cleanroom hook-up achieved 100% tool-up-time targets and zero audit rework to IEC 60204/61511/ISO 14644.

CORE COMPETENCIES

Electrical Power Systems

LV/MV/HV switchgear · Power distribution · Protection coordination · Short-circuit, load-flow & arc-flash studies (ETAP) · VFD/Variable Frequency Drive systems · Technical bid evaluation · Basis of Design · Pre-FEED / FEED / Detailed Engineering · MV 11 kV systems

Marine & Offshore

FPSO topsides · Jack-up drilling rigs · Offshore wind substations · Hybrid fast crew boats (FCB) · Crew Transfer Vessels (CTV) · Ro-Pax passenger ferries · Autonomous/electric tugs · Sea trials & sail-away · EPC project delivery · Concept-to-handover lifecycle · Offshore converter platforms (emerging)

Hybrid, Renewable & Energy Storage

Li-ion BESS (72 kWh commissioned) · BMS · DC UPS · DC/DC inverters · Shaft generators · Solar-wind hybrid · Shore power / cold ironing · IEC 62619 battery-safety (in progress) · IEC 61660 DC/LVDC zonal distribution (in progress)

Automation, Controls & Digital

PLC · SCADA (ABB 800xA, Siemens, Yokogawa) · IEC 61850 · PMS · IAS · DP1/DP2 · Integrated bridge & navigation · Fire & Gas · Python engineering automation · AI agent workflow integration · IACS E26/E27 OT cybersecurity (in progress) · IEC 62443

Compliance, Standards & Class

IEC 60092 · IEC 61892 · IEC 60364 · IECEx/ATEX Zone 1/2 · SOLAS Ch. II-1 · IMO MSC · MARPOL · DNV · ABS · LR · BV · NSCV · ISO 9001/50001 Lead Auditor · FAT/SAT · FMEA · Cause & effect · Hazardous-area Ex equipment selection

Core Tools

ETAP · SKM Power Tools · Math CAD · MATLAB/Simulink · AutoCAD Electrical · ABB 800xA · Siemens/Yokogawa SCADA · Fluke DSX2-8000 · Python (engineering automation)

PROFESSIONAL EXPERIENCE

Electrical Engineer — Hybrid Marine Power Jul 2024 – Present

Strategic Marine Pte. Ltd. · Marine / Newbuild · Singapore

Architected the complete LV power system for a 42 m Gen-4 Fast Crew Boat ($3 \times$ CAT C32; 3×150 kW; VEEM propulsion) — SLDs, load lists, cable schedules — securing first-pass class approval across a programme portfolio spanning ABS, BV, LR & DNV.

Integrated a 72 kWh Li-ion BESS, BMS, DC/DC inverters and 90 kW shaft generator under LR Hybrid Power notation, cutting fuel burn by up to 50% in low-load and loitering modes; led BMS-to-PMS interface and FAT acceptance with LR surveyor.

Delivered full electrical design (SLDs, AMS I/O, cable schedules, IEC 60092-compliant IT earthing) across a multi-hull Gen-4 FCB programme, offshore-wind CTV series, 58 m Ro-Pax and 43 m fast ferry — dual 380/220 V AC + 24 V DC, interlocked shore power to BV/LR/AMSA-NSCV.

FAT-validated ABB azimuth propulsion PMS on an autonomous-electric tug; led cross-fleet cathodic-protection and stray-current engineering; standardised departmental deliverables, KPI tracking and basis of design documentation.

Senior Project Engineer — Semiconductor & High-Tech Feb 2023 – Jul 2024

Exyte · Semiconductor / Cleanroom · Singapore

Led 300 mm semiconductor cleanroom power hook-up to IEC 60204/61511/ISO 14644 — zero audit rework against 100% tool-uptime targets; PLC/SCADA interlock and functional-safety validation cleared first pass.

Introduced prefabricated power skids that cut on-site labour by ~50% and eliminated contamination risk in a live cleanroom environment.

Drove ISO 50001 energy stewardship through smart load monitoring, measurably reducing fab energy intensity; standardised site documentation as certified Lead Auditor.

E&I Engineer — Hazardous-Area, Renewables & Power Jan 2021 – Jan 2023

Phoenix Meccano (S.E.A.) Pte Ltd · Industrial / Offshore · Singapore

Pioneered Malaysia's first solar-wind autonomous unmanned wellhead platform (24 V DC solar/battery bus) — eliminated routine diesel use and enabled full remote operation, with zero unplanned shutdowns at commissioning.

Directed an onshore cogeneration plant E&I scope — transformers, MV/LV switchgear, 400 kVA UPS, SLD, tie-in and instrument-loop design; ABB/Siemens/Yokogawa SCADA integration and technical bid evaluation for all major E&I packages.

Designed IECEx/ATEX Zone 1/2 hazardous-area systems — Ex equipment selection, IS barriers, zoning studies — and standardised compliance documentation as certified Lead Auditor (ISO 9001 & ISO 50001).

E&I Superintendent — FPSO, Rigs & Offshore Wind Nov 2012 – Dec 2020

Keppel FELS (Keppel Offshore & Marine) · Offshore / Marine · Singapore

Superintended crews of up to 100+ across fabrication, installation and integrated commissioning of FPSO topsides (Pre-FEED through detailed engineering and handover) — energised disconnectable turret, MV 11 kV HV switchgear and ABB 800xA Fire & Gas; FPSO MV/LV reliability maintained at 100% uptime through sail-away.

Energised an offshore wind substation topside to 100% IEC 61850 conformity — full protection-relay coordination, alarm validation and sail-away sign-off; zero punch-list items outstanding at class handover.

Commissioned ultra-deepwater jack-up drilling rigs and semisubmersible accommodation modules through FMEA, cause-and-effect matrices, FAT/SAT and sea trials; mentored 15+ engineers via Innovation & Quality Circles.

Authored 30+ E&I risk assessments; delivered integrated bridge & navigation systems (gyro, VDR, alarm & watch systems); led EPC detailed-engineering deliverables for multiple concurrent vessel builds.

Assistant Electrical Engineer — Industrial & Infrastructure Projects Sep 2009 – Oct 2012

Ananda Shipyard & Slipways Ltd · Bangladesh / APAC

Led E&I crews of 15–30 on newbuild vessels and rigs — commissioned Azimuth Thrusters, Controllable Pitch Propellers, and bridge navigation equipment to ABS / GL class and owner acceptance, achieving first-time milestone sign-off.

Validated cable routing and interconnection designs against load schedules and voltage-drop criteria before energisation; governed BOM and delivery scheduling to protect on-site readiness against supply-chain disruption

Delivered LV/MV power distribution, protection coordination and commissioning across industrial and infrastructure projects — establishing the foundation of 16 years' uninterrupted career in electrical engineering.

EDUCATION & PROFESSIONAL CREDENTIALS

EDUCATION

MSc Sustainable Electricity Generation — Universidad Católica de Murcia (UCAM), Spain — Apr 2026

MSc Digital Transformation — London Metropolitan University, UK — Mar 2026

Master of Business Administration (MBA) — University of Sunderland, UK — Jun 2025

BSc Electrical & Electronic Engineering — Bangladesh University — Jul 2008

Diploma in Electronics Engineering — Jessore Polytechnic Institute — Jan 2004

CERTIFICATIONS & TRAINING

IECEx Ex 001 — Basic Principles of Protection in Explosive Atmospheres (valid to Oct 2027)

ISO 50001:2018 Energy Management — Lead Auditor (CQI IRCA)

ISO 9001:2015 Quality Management — Lead Auditor (CQI IRCA)

Shipyard Supervisors Safety Course (SSSC)

WSQ WSH Management in Construction Industry (formerly BCSS)

Roxtec Regular System Installer | CMP Cable Gland Installer | RAYCHEM APKT/EPKT

ABB — KFELS B-Class Jack-up Drilling Rig LV Switchboard Training

Fluke DSX2-8000 Cable Analyzer Copper Certifier Training

Work at Height (Supervisor), Confined Space Safety — Keppel Offshore & Marine

WSH in Process Plant — Eversafe Academy

EMERGING & FUTURE-READY CAPABILITIES

Actively developing the next generation of marine-electrical expertise to lead the energy transition: IACS E26/E27 OT cybersecurity (IEC 62443) · Shore power / cold ironing (IEC/ISO/IEEE 80005-1) · DC/LVDC zonal distribution (IEC 61660) · IEC 62619 Li-ion battery safety certification · Offshore converter platforms · Python engineering automation & AI agent workflow integration.